

Teste Abrangente de Formulas Matematicas — Typst

1. Aritmetica Basica e Fracoese

$$\begin{aligned}a + b &= c \\ a - b &= c \\ a \times b &= c \\ a \cdot b &= c \\ \frac{a}{b} &= c \\ \frac{a + b}{c + d} &= e \\ a^2 + b^2 &= c^2 \\ \sqrt{a^2 + b^2} &= c \\ \sqrt[3]{x} &= y\end{aligned}$$

2. Simbolos Gregos

$$\begin{aligned}\alpha + \beta &= \gamma \\ \mathsf{A} + \mathsf{B} &= \Gamma \\ \delta, \Delta, \varepsilon, \epsilon \\ \zeta, \eta, \theta, \vartheta, \iota, \kappa \\ \lambda, \Lambda, \mu, \nu, \xi, \Xi \\ \pi, \Pi, \rho, \varrho, \sigma, \Sigma, \varsigma \\ \tau, \upsilon, \Upsilon, \varphi, \phi, \Phi \\ \chi, \psi, \Psi, \omega, \Omega\end{aligned}$$

3. Relacoes e Operadores Logicos

$$\begin{aligned}a < b \\ a > b \\ a \leq b \\ a \geq b \\ a \neq b \\ a \approx b \\ a \equiv b \\ a \propto b \\ a \in B \\ a \subset B \\ a \subseteq B \\ a \supset B \\ a \supseteq B \\ a \cup B \\ a \cap B \\ \varnothing \\ \forall x \in \mathbb{R} \\ \exists y \in \mathbb{N} \\ \neg(a = b) \\ a \wedge b \\ a \vee b \\ a \Rightarrow b \\ a \Leftrightarrow b\end{aligned}$$

4. Calculo e Analise

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}f(x) &= f'(x) \\ \frac{\mathrm{d}^2y}{\mathrm{d}x^2} \\ \int_a^b f(x)\mathrm{d}x \\ \int_0^\infty e^{-x}\mathrm{d}x &= 1 \\ \oint_C f(z)\mathrm{d}z = 2\pi i \\ \sum_{k=1}^n k = \frac{n(n+1)}{2} \\ \prod_{k=1}^n k &= n! \\ \lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1 \\ \lim_{x \rightarrow \infty} \frac{1}{x} &= 0 \\ \frac{\partial}{\partial x}f(x, y) \\ \nabla f(x, y, z)\end{aligned}$$

5. Algebra Linear

$$\begin{aligned}\begin{pmatrix}1 & 2 \\ 3 & 4\end{pmatrix} \\ \begin{pmatrix}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{pmatrix} \\ \begin{pmatrix}1 \\ 2 \\ 3\end{pmatrix} \\ \begin{bmatrix}1 \\ 2 \\ 3\end{bmatrix} \\ \det\begin{pmatrix}1 & 2 \\ 3 & 4\end{pmatrix} \\ A^T \\ A^{-1} \\ A^\dagger \\ \mathrm{tr}(A) \\ \mathrm{rank}(A) \\ \dim(V) \\ \ker(A) \\ \mathrm{im}(A) \\ \langle u, v \rangle \\ \|v\| \\ |x| \\ \lfloor x \rfloor \\ \lceil x \rceil\end{aligned}$$

6. Funcoes Especiais

$$\begin{aligned}\sin x, \cos x, \tan x \\ \arcsin x, \arccos x, \arctan x \\ \sinh x, \cosh x, \tanh x \\ \log_a x, \ln x, \exp x \\ \max(a, \; b), \min(a, \; b) \\ \gcd(a, \; b), \mathrm{lcm}(a, \; b) \\ \lfloor x \rfloor, \lceil x \rceil, \llbracket x \rrbracket \\ \mathrm{sgn}(x) \\ \mathrm{erf}(x) \\ \mathrm{Res}(f(z), z = z_0)\end{aligned}$$

7. Conjuntos e Teoria dos Numeros

$$\begin{aligned}\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C} \\ \{x \in \mathbb{R} \mid x > 0\} \\ \mathcal{P}(A) \\ \aleph_0 \\ \beth_1 \\ \binom{n}{k} \\ n! = n \cdot (n-1) \cdot \ldots \cdot 1 \\ \binom{n}{k} \\ k) = \frac{n!}{k!(n-k)!}\end{aligned}$$

8. Alinhamento de Equacoes

$$\sum_{k=0}^n k = 1 + \ldots + n \setminus = \frac{n(n+1)}{2} \setminus = \frac{n^2}{2} + \frac{n}{2}$$
$$\frac{3x+y}{7} = 9 \text{ dado } \setminus 3x + y = 63 \text{ multiplique por } 7 \setminus 3x = 63 - y \text{ subtraia y } \setminus x = 21 - \frac{y}{3} \text{ divida por } 3$$

9. Funcoes por Partes e Casos

$$\begin{aligned}f(x) &= \begin{cases} 0 & \text{se } x < 0 \\ x^2 & \text{se } 0 \leq x < 1 \\ 1 & \text{se } x \geq 1 \end{cases} \\ | \; x \; | &= \begin{cases} x & \text{se } x \geq 0 \\ -x & \text{se } x < 0 \end{cases}\end{aligned}$$

10. Decoradores e Acentos

$$\begin{aligned}\hat{x} \\ \tilde{x} \\ |x| \\ (x) \\ \dot{x} \\ \dot{\dot{x}} \\ \frac{a+b+c}{\text{soma}} \\ \overbrace{a+b+c}^{\text{soma}} \\ \overline{a+b+c} \\ \overline{\overline{a+b+c}} \\ a\!\!\!\diagup\!\!\!b \\ a\!\!\!\diagdown\!\!\!b\end{aligned}$$

11. Fisica e Quimica

$$\begin{aligned}E = mc^2 \\ F = G\frac{m_1m_2}{r^2} \\ p = mv \\ L = r \times p \\ H\psi = E\psi \\ \nabla \cdot E = \frac{\rho}{\varepsilon_0} \\ \nabla \times B = \mu_0 J + \mu_0 \varepsilon_0 \frac{\partial E}{\partial t} \\ \mathrm{d}S \geq \frac{\delta q}{T} \\ pV = nRT \\ \lambda = \frac{h}{p}\end{aligned}$$

12. Probabilidade e Estatistica

$$\begin{aligned}P(A \mid B) &= \frac{P(B \mid A)P(A)}{P(B)} \\ E[X] &= \sum_i x_i P(X = x_i) \\ \mathrm{Var}(X) &= E[X^2] - (E[X])^2 \\ \sigma &= \sqrt{\mathrm{Var}(X)} \\ X &\sim N(\mu, \sigma^2) \\ \Phi(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} \mathrm{d}t \\ \hat{\mu} &= \frac{1}{n} \sum_{i=1}^n X_i\end{aligned}$$

13. Geometria e Topologia

$$\begin{aligned}\pi r^2 \\ 2\pi r \\ \frac{4}{3}\pi r^3 \\ d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \\ d(x, y) = \max_i |x_i - y_i| \\ d(x, y) = \sum_{i=1}^n |x_i - y_i| \\ \chi(M) = V - E + F\end{aligned}$$

14. Numeros Complexos

$$\begin{aligned}i^2 &= -1 \\ z &= a + bi \\ | \; z \; | &= \sqrt{a^2 + b^2} \\ z^* &= a - bi \\ e^{i\theta} &= \cos \theta + i \sin \theta \\ e^{i\pi} + 1 &= 0\end{aligned}$$

15. Series e Transformadas

$$\begin{aligned}f(x) &= \frac{a_0}{2} + \sum_{n=1}^\infty (a_n \cos(nx) + b_n \sin(nx)) \\ F(\omega) &= \int_{-\infty}^\infty f(t) e^{-i\omega t} \mathrm{d}t \\ f(t) &= \frac{1}{2\pi} \int_{-\infty}^\infty F(\omega) e^{i\omega t} \mathrm{d}\omega \\ \sum_{n=0}^\infty x^n &= \frac{1}{1-x} \text{ para } |x| < 1\end{aligned}$$

16. Fontes e Estilos Especiais

$$\begin{aligned}\mathbb{R} \\ \mathcal{L} \\ \mathfrak{G} \\ \code \\ \text{text} \\ \text{sans} \\ \mathbf{x} \\ \boldsymbol{x} \\ x\end{aligned}$$

17. Subscritos e Sobrescritos Complexos

$$\begin{aligned}x_{i,j}^{k,l} \\ \sum_{i=1}^n \sum_{j=1}^m a_{ij} \\ \int_V f(x) \mathrm{d}V \\ \bigcup_{i \in I} A_i \\ \bigcap_{i \in I} A_i \\ \infty \\ \aleph \\ \beth\end{aligned}$$

18. Expressoes Aninhadas

$$\begin{aligned}\frac{1}{1 + \frac{1}{1 + x}} \\ \sqrt{a + \sqrt{b + \sqrt{c}}} \\ \frac{e^{x^2+y^2}}{x^2+y^2} \\ \frac{\partial^2 f}{\partial x \partial y} \\ \overbrace{\frac{a+b}{2}}^{\text{top}} \\ \underbrace{\frac{a+b}{2}}_{\text{bottom}}\end{aligned}$$

19. Espacamento e Tamanhos

$$\begin{aligned}a \; b \\ a \; \! b \\ a \; \; b \\ a \; \; \; b \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2}\end{aligned}$$

20. Referencias Cruzadas e Numeracao

$$\begin{aligned}E = mc^2 \quad (1) \\ F = ma \quad (2) \\ \left[\hat{x}, \hat{p} \right] = i \; \hbar \quad (25) \\ i \; \hbar \frac{\partial \Psi}{\partial t} = \hat{H} \Psi \quad (26) \\ \Psi(x, \; t) = \sum_n c_n \psi_n(x) e^{-iE_n \frac{t}{\hbar}} \quad (27) \\ \langle \varphi | | \psi \rangle = \int \varphi^*(x) \psi(x) \mathrm{d}x \quad (28) \\ \langle A^* \rangle = \langle \psi | \hat{A} | \psi \rangle \quad (29)\end{aligned}$$

27. Relatividade

$$\begin{aligned}\mathrm{d}s^2 = -c^2 \mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 \quad (30) \\ \Gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (31) \\ E^2 = (pc)^2 + (mc^2)^2 \quad (32) \\ R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (33)\end{aligned}$$

28. Teoria dos Grafos

$$\begin{aligned}\chi(G) \geq \omega(G) \quad (34) \\ \alpha(G) + \tau(G) = | \; V(G) \; | \quad (35) \\ \sum_{v \in V(G)} \deg(v) = 2 \mid E(G) \mid \quad (36) \\ K_n, C_n, P_n, Q_n \quad (37)\end{aligned}$$

29. Teoria da Informacao

$$\begin{aligned}H(X) &= - \sum_{x \in X} P(x) \log_2 P(x) \quad (38) \\ I(X; Y) &= H(X) - H(X \mid Y) \quad (39) \\ D_{KL}(P \mid Q) &= \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (40) \\ C &= \max_{P_X} I(X; Y) \quad (41)\end{aligned}$$

30. Otimizacao

$$\begin{aligned}\min_{x \in \mathbb{R}^n} f(x) \text{ sujeito a } g_{i(x)} \leq 0, h_{j(x)} = 0 \quad (42) \\ L(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_{i(x)} + \sum_j \mu_j h_{j(x)} \quad (43) \\ \nabla f(x^*) + \sum_i \lambda_i^* \nabla g_{i(x^*)} + \sum_j \mu_j^* \nabla h_{j(x^*)} = 0 \quad (44)\end{aligned}$$